NAVIN CHANDRA

Curriculum Vitae

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Education

2025 (May) **PhD, Physical Oceanography & Ocean Modelling**, *Centre for Atmospheric Sciences, Indian Institute of Technology Delhi*, New Delhi, India.
Study of Oceanic Mesoscale Eddies in the Bay of Bengal using a Numerical Model.

2016 : **Master of Technology, Atmospheric & Oceanic Science and Technology**, *Centre for Atmospheric Sciences, Indian Institute of Technology Delhi*, New Delhi, India.

2012 : **Bachelor of Technology, Mechanical Engineering**, *Biju Pattnaik University of Technology*, Rourkela, India.

Work Experience

National Centre for Medium Range Weather Forecasting (NCMRWF)

March,2023 – present ***Project Scientist-II***.
Currently I am working as a Project Scientist II, in Bimstec Centre for Weather and Climate (BCWC) and look after weather system of BIMSTEC memeber countries.

Indian Institute of Technology Delhi

Nov,2016 – June 2017 ***Project Assistant***.
Visualization and analysis of different model output

Project Incharge : **Prof. Vimlesh Pant**, *Professor, Centre for Atmospheric Sciences*, IIT Delhi (*Personal Web-page*)

Publications

Journal Articles

2025 **Navin Chandra**, Vimlesh Pant, and Kumar Ravi Prakash. Influence of cyclonic and anticyclonic mesoscale oceanic eddies on thermodynamic characteristics of near-surface atmospheric column. *Quarterly Journal of the Royal Meteorological Society*, 2025. `https://doi.org/10.1002/qj.5036`.

2025 **Navin Chandra** and Vimlesh Pant. Variability of mesoscale eddies and their kinetic energy in the bay of bengal using a high-resolution ocean model. *Regional Studies in Marine Science.*, volume 89, 2025. `https://doi.org/10.1016/j.rsma.2025.104329`.

2025 **Navin Chandra** and Vimlesh Pant. Impact of indian ocean dipole on the mesoscale eddies and their energy in the bay of bengal. *Oceanologia*, volume 67(1), 2025. `https://doi.org/10.5697/ygvi7171`.

2022 Kumar Ravi Prakash, Vimlesh Pant, T. V. S. Udaya Bhaskar, and **Navin Chandra**. What made the sustained intensification of tropical cyclone fani in the bay of bengal? an investigation using coupled atmosphere–ocean model. *Atmosphere*, volume 13, 2022. `https://doi.org/10.3390/atmos13040535`.

2021 Kumar Ravi Prakash, Tanuja Nigam, Vimlesh Pant, and **Navin Chandra**. On the interaction of mesoscale eddies and a tropical cyclone in the bay of bengal. *Natural Hazards*, volume 106, 2021. `https://doi.org/10.1007/s11069-021-04524-z`.

Communicated/Under Preparation Article

2026 **Navin Chandra**, *and Vimlesh Pant* , Nonlinear Eddy Response to Wind-Stress Amplitude in the Bay of Bengal: Energy Saturation and a Shift Toward Anticyclone Dominance, For *Under Preparation for JGR:Ocean*.

2026 **Navin Chandra**, , *Vimlesh Pant and Imranali M. Momin*, Where the Cascade Breaks: How Ocean Model Resolution Silently Erases the Ageostrophic Energy Pathway in the Bay of Bengal, *Journal of Physical Oceanography*.
Under Review

2026 **Navin Chandra**, *Ankur Gupta, Akhilesh Kumar Mishra and Raghavendra Ashrit*, Eddy activity on both sides of the Indian Peninsula with the help of the gridded sea surface height, In *Journal of Oceanography*.
Under Review

2026 **Navin Chandra**, *Jivesh Dixit, Imranali M. Momin and Raghavendra Ashrit*, Quantifying Interpolation-Induced Representativeness Error in Satellite SST over the Indian Ocean, In *Quarterly Journal of the Royal Meteorological Society*.
Under Review

Report

2026 **Navin, Chandra**, Ankur Gupta, Akhilesh Kumar Mishra, and Raghavendra Ashrit. Location based marine forecast of major port cities of india and BIMSTEC member countries, 2026. `https://doi.org/10.64349/nmrf.tr.4.2026`.

2024 **Navin Chandra**, Mohan S Thota, and Raghavendra Ashrit. Trend analysis of rainfall, maximum temperature, and minimum temperature over thailand using CRU TS 4.07 monthly data, 2024. `https://doi.org/10.64349/nmrf.rr.5.2024`.

2024 Gibies George John P. George Sumit Kumar Laxmikant Dhage Nishtha Agarwal Durgesh Nandan Piyush **Navin Chandra** Tanvi Gupta, S. Indira Rani and V. S. Prasad. Microsat-2b (eos-07) microwave humidity sounder radiance assimilation in the ncum system, 2024. `https://doi.org/10.64349/nmrf.rr.2.2024`.

Book Chapter

2026 , *Sankar Prasad Lahiri*, **Navin Chandra**, Sauvik Santra, Deepak Singh Bisht and Vimlesh Pant, Downscaling Projections from Global Climate Models for Planning Disaster Resilient National Highways.
Under Review

In Conference Proceedings

2022 **Navin Chandra** and Vimlesh Pant. Mesoscale eddies in the bay of bengal and their response to iod modes. In AGU General Assembly, *held during 12-16 Dec 2022 (online)*, 2022.

2021 **Navin Chandra** and Vimlesh Pant. Variability of eddy kinetic energy in the bay of bengal in the pretext of indian ocean dipole. In AGU General Assembly, *held during 13-17 Dec 2021 (online)*, 2021.

2021 **Navin Chandra** and Vimlesh Pant. Characteristics of mesoscale eddies in the bay of bengal using a numerical model. In OSICON-2021, *held during 12-14 August 2021 (online)*, 2021.

2020 **Navin Chandra** and Vimlesh Pant. Eddies and their energetics in the bay of bengal. In EGU General Assembly, *held during 4-8 May 2020 (online)*, 2020.

2016 **Navin Chandra**, K.K. Sandeep, and Vimlesh Pant. Simulation of sea surface variables over the bay of bengal in the high resolution eddy-resolving set-up of roms model. In Recent Advancement in Regional Ocean Modelling (RAROM-2016)., *CORAL-Indian Institute of Technology Kharagpur, March-2016.*, 2016.

Fellowships & Awards

2016 **Appreciation for Master's Research Work** by *IIT DELHI ALUMNI ASSOCIATION RESEARCH & INNOVATION AWARD*

2014 –2016 *Fellowship from Ministry of Human Resource and Development(MHRD)*, Government of India for master studies

2017 –2022 *Fellowship from Ministry of Human Resource and Development(MHRD)*, Government of India for PhD research.

Technical skills

Numerical Model	ROMS, COAWST
Programming Languages	Python, Fortran, Matlab, Shell scripting
Operating System	Windows, Linux, macOS
Other Tool	CDO, NCO, Ferret, Git

SERVICE AND OUTREACH

Reviewer Regional Studies in Marine Sciences, Theoretical and Applied Climatology, Environmental Monitoring and Assessment, Scientifc Data

2025-present

Referees

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